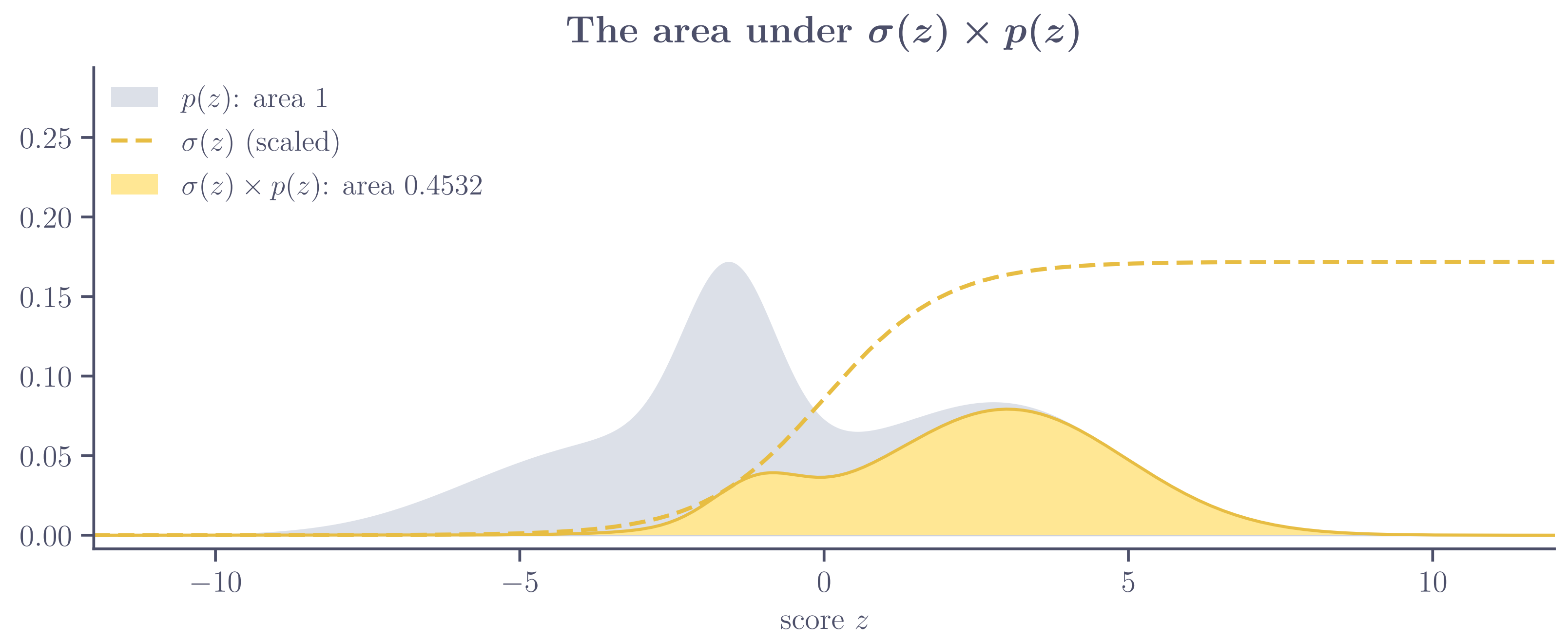
GOAL

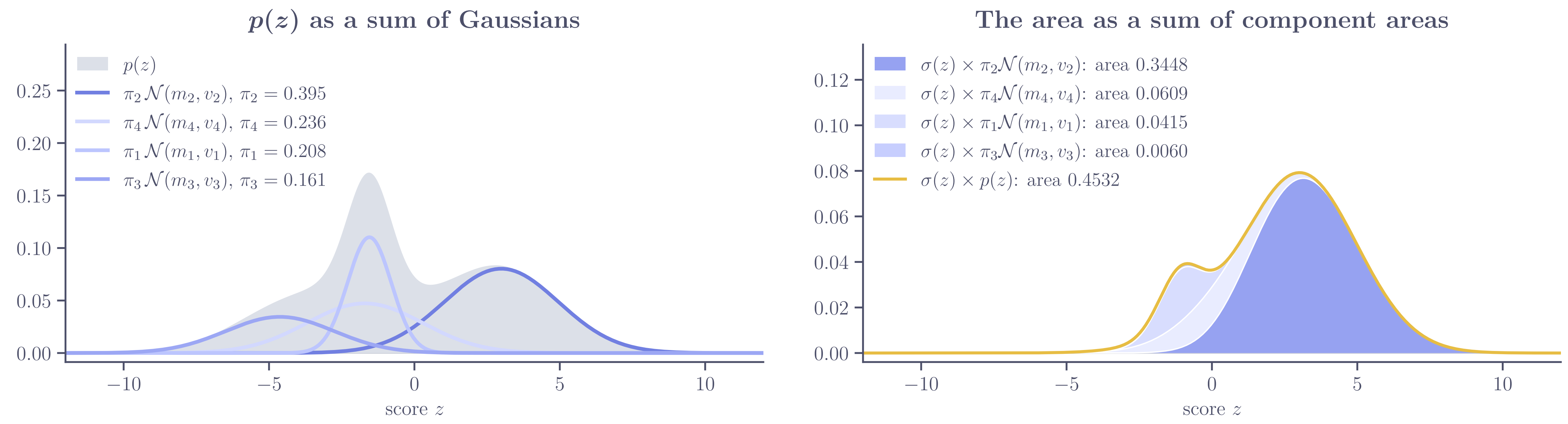
What we want to compute


The area under $\sigma(z) \times p(z)$

Some features are missing, so the score $z = w^\top x$ is uncertain: $p(z)$, from the density model (grey, area 1). The expected prediction $\mathbb{E}[\sigma(z) \mid x_{\text{obs}}]$ is the yellow area under $\sigma(z) \times p(z)$. For a Gaussian mixture model this area can be computed directly, with one small quadrature per component.

STEP 1

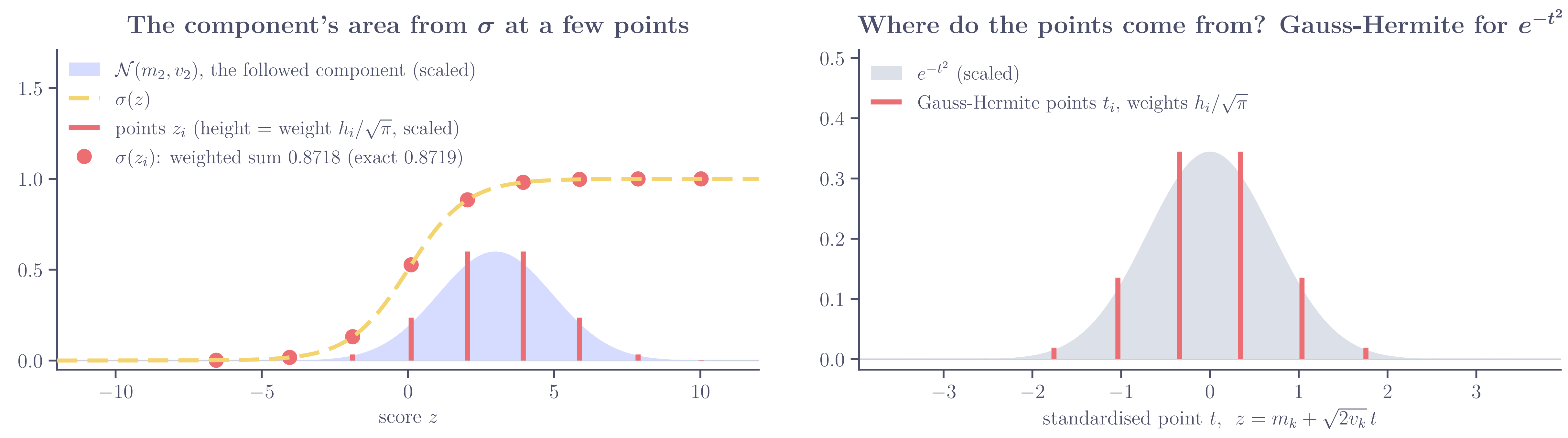
$p(z)$ is a mixture of Gaussians, so the area splits by component


$p(z)$ as a sum of Gaussians — **The area as a sum of component areas**

For the GMM, z given the observed features is again a mixture: in component k it is Gaussian, $z \mid k \sim \mathcal{N}(m_k, v_k)$, with weight $\pi_k = p(k \mid x_{\text{obs}})$, mean $m_k = w^\top \mu_k$ (observed features filled in) and variance $v_k = \sum_{j\,\text{missing}} w_j^2 \sigma_{kj}^2$. So the area splits into one area per component: $\mathbb{E}[\sigma(z)] = \sum_k \pi_k \, \mathbb{E}[\sigma(z) \mid k]$, and each $\mathbb{E}[\sigma(z) \mid k]$ is a 1D integral of σ against a Gaussian. The dark indigo component (largest weight) is followed in the next steps.

STEP 2

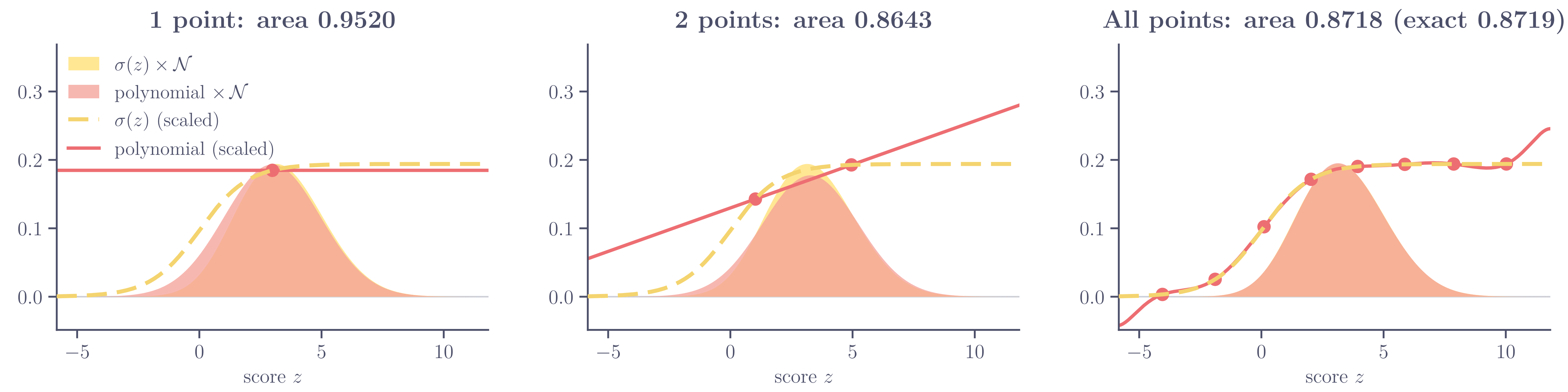
The area of a component is a weighted sum of σ at a few points


The component's area from σ at a few points — **Where do the points come from? Gauss-Hermite for e^{-t^2}**

Gauss-Hermite replaces the integral of $\sigma(z)$ against the Gaussian of the component by a weighted sum $\sum_i \frac{h_i}{\sqrt{\pi}} \sigma(z_i)$ over 10 points z_i (lines, height = weight). Only σ is evaluated, the Gaussian enters through where the points are and how much they weigh. The points t_i and weights h_i are fixed once, for the weight e^{-t^2} (grey), and moved to every component by $z_i = m_k + \sqrt{2v_k}\,t_i$. They are chosen so that the sum is exact for every polynomial up to degree $2n-1$: the points crowd where the Gaussian has its mass.

STEP 3

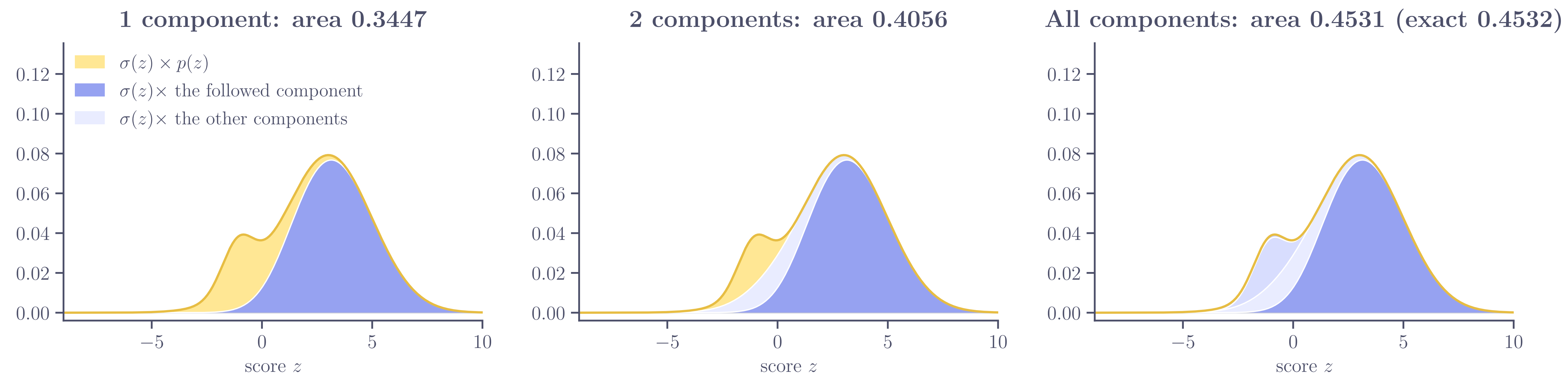
Why it works: the sum is the area under a polynomial through σ


1 point: area 0.9520 — **2 points: area 0.8643** — **All points: area 0.8718 (exact 0.8719)**

Every panel: with n points the sum is exactly the area under the polynomial of degree $n-1$ through σ at the points (coral line and dots, scaled) times the Gaussian of the component (coral area), against $\sigma(z)$ (yellow, dashed) and its area. One point is a constant, two a line. As the polynomial closes in on σ where the Gaussian has its mass, the area closes in on the component's area. Far out the polynomial runs off, but there the Gaussian is about 0.

STEP 4

The component areas add up to the area under $\sigma(z) \times p(z)$


1 component: area 0.3447 — **2 components: area 0.4056** — **All components: area 0.4531 (exact 0.4532)**

Every panel: $\sigma(z)$ times the sum of the components with the largest weights (dark indigo, then lighter, stacked) and its area $\sum_k \pi_k \, \mathbb{E}[\sigma(z) \mid k]$, every component area from 10 Gauss-Hermite points, against the goal (yellow). With all components it is the GMM Gauss-Hermite estimate 0.4531 (exact 0.4532).